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## Matematické metody v ekonomii

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# THE DOT-COM BUBBLE AND ITS CONSEQUENCES FOR THE GULF COUNTRIES

### 1.1. The nature of the crisis and shock transmission channels

The period of 2000–2002 was marked by a major financial crisis known in economic literature as the "Dot-com Bubble." This phenomenon represented a period of speculative frenzy surrounding internet companies on the stock markets of the United States and other developed countries during 1995–2000, characterized by asset valuation based on hypothetical "potential" rather than fundamental financial indicators (such as profits or revenues) [1, 2]. The bubble peaked in March 2000, when the NASDAQ index reached a historical high of 5,048 points, followed by a prolonged collapse: by October 2002, the index had lost approximately 78% of its value [3].

Unlike subsequent global shocks, the dot-com crisis did not lead to a systemic collapse of the global banking system, which limited its direct transmission to emerging markets outside the technology sector. For countries with dominant oil and gas economies, including all member states of the Gulf Cooperation Council (GCC), the primary impact of the shock was transmitted not through financial channels, but through the dynamics of global industrial demand and energy prices.

**Mechanisms of Global Shock Transmission to GCC Economies**

Although the Gulf countries were not direct participants in the technology boom, their macroeconomic stability came under threat through four interconnected channels of external shock transmission. Unlike the 2008 crisis, when the collapse in global capital markets led to a sharp widening of sovereign debt spreads in GCC countries and outflow of foreign capital, during the dot-com crisis the financial transmission

channels remained weak: the region's banking systems remained isolated from the global technology bubble, and the share of foreign portfolio investments in local market assets was minimal. The transmission channels of the "dot-com bubble" crisis into the economy, using Oman as an example, are presented below in Section 1.2.

### Comparative Analysis of Global Shocks: 2000, 2008, and 2020

To understand the scale and specificity of the "dot-com bubble" crisis, it is appropriate to conduct a comparison with subsequent global upheavals. As demonstrated by the data in Table 1, each of these crises had a unique nature, but all of them significantly affected the economies of GCC countries through energy demand channels.

**Table 1 Comparative Characteristics of Three Global Economic Shocks (2000–2020)**

| No. | Parameter | Dot-com Bubble (2000–2002) | Mortgage Crisis (2008–2009) | Pandemic Shock (2020) |
|---|---|---|---|---|
| 1 | Epicenter | USA, technology sector | USA, financial sector | Global, services sector |
| 2 | Trigger | Overvaluation of internet startups, NASDAQ collapse | Subprime mortgage collapse, Lehman Brothers bankruptcy | COVID-19 pandemic + OPEC+ price war |
| 3 | Peak → Collapse | March 2000 → October 2002 (–78% NASDAQ) | September 2008 → March 2009 (–57% S&P 500) | February 2020 → March 2020 (–34% global GDP) |
| 4 | Impact on oil prices | –30% (from $28 to $20/barrel) | –75% (from $145 to $35/barrel) | –65% (from $65 to $20/barrel) |

| | | | | |
|---|---|---|---|---|
| 5 | Impact on Oman's GDP | Slowdown to 0.7% (2002) | –5.2% real GDP (2009) | –2.8% real GDP (2020) |
| 6 | Oman's budget deficit | ~2–3% of GDP | ~5% of GDP | ~16–18% of GDP |
| 7 | Monetary response of the Central Bank of Oman | Liquidity maintenance, gradual rate cuts | Bank support program, liquidity injections | TESS ($8.5 billion), rate cuts to 0.25% |
| 8 | Fiscal response | Spending restraint, use of reserves | Anti-crisis budget, increased social spending | $12 billion stimulus package, tax deferrals |

**Sources**: compiled based on calculated data [1, 3, 5, 6, 9, 10].

### 1.2. Detailed Analysis of GCC Countries' Response to the 2001 Shock

**Bahrain's Response: Lowest Oil Dependence Due to Diversification**

Bahrain, possessing the most modest hydrocarbon reserves among GCC countries, historically had to rely on non-oil sectors. By 2000, financial services, the aluminum industry (Alba), and tourism constituted a significant share of GDP and tax revenues [6, 11]. As a result, the decline in oil and gas budget revenues in 2001 was the smallest in the region—only about –15% [7, 8]. Bahrain's real GDP maintained stable growth rates (~4.0%), and the government even had limited fiscal space to support investment projects [5]. The crisis confirmed the correctness of the diversification strategy, and in subsequent years Bahrain continued to develop its financial hub and industrial zones.

**Bahrain's Lesson:** An economy with an initially low share of oil rent and a developed private sector proves to be least sensitive to commodity shocks, making diversification the most effective long-term strategy for small hydrocarbon exporters.

**Kuwait's Response: The Role of the Sovereign Fund as a Stabilizer**

Kuwait, with one of the highest shares of oil revenues in the budget in the region, faced a decline in oil and gas revenues of approximately –24% in 2001 [6, 7]. However, thanks to the existence of the Future Generations Fund, the government was able to compensate for the lost revenues through withdrawals from accumulated reserves, which allowed it to maintain positive real GDP growth rates of ~1.5% [5, 9]. Unlike Oman, Kuwait did not pursue a path of strict fiscal consolidation, which prevented a contraction in domestic demand. At the same time, the crisis stimulated a review of budgetary policy: in subsequent years, budgetary regulation rules were strengthened, and allocations to reserve funds were increased.

**Kuwait's Lesson:** The presence of a sovereign wealth fund managed according to clear withdrawal rules allows separating budgetary consumption from short-term oil price volatility, maintaining macroeconomic stability even with a significant decline in revenues.

**Oman: Vulnerability of an Economy Without a Stabilization Fund**

Oman proved to be the most vulnerable economy among GCC countries to the 2001 crisis. Despite an identical decline in global oil prices (~–27%), the reduction in the Sultanate's budget revenues from oil and gas was approximately –26%, the highest figure in the region [1, 4, 6]. The main reason was the absence at that time of a large sovereign stabilization fund that could compensate for lost revenues. As a result, the government was forced to resort to strict fiscal consolidation—direct cuts in government spending—which directly hit aggregate demand and the real sector [2, 5].

Oman's real GDP in 2002 slowed (growth rate) to 0.7%, whereas in previous years growth had been 4–5%. Oman’s real GDP growth slowed to 0.7% in 2002, whereas in the preceding years the growth rate stood at 4–5%, as shown in Table 2.

At the same time, monetary authorities limited themselves to maintaining liquidity in the banking system and a gradual reduction of interest rates following the US Federal Reserve, but the fiscal channel proved to be dominant. The 2001 crisis became a "hard lesson" for Oman: it subsequently stimulated the creation of the State

General Reserve Fund and the launch of the long-term diversification program Oman Vision 2040 [2, 4].

**Table 2. Transmission Channels of the "Dot-com Bubble" Crisis into Oman's Economy**

| No. | Transmission Stage | Impact Mechanism | Quantitative Estimate | Source |
|---|---|---|---|---|
| 1 | Global demand | Slowdown of the global economy → decline in energy consumption | –3.2% (global oil demand, 2001) | [3], IEA |
| 2 | Price channel | Decline in export hydrocarbon prices | –27% (Oman Crude, 2001 vs. 2000) | [6], Central Bank of Oman |
| 3 | Budget channel | Reduction in oil and gas revenues to the treasury | –26% (budget revenues, 2001) | [4], estimate based on Central Bank data |
| 4 | Real sector | Fiscal consolidation → decline in aggregate demand | GDP growth at 0.7% (2002) | [5], World Bank |

**Note:** compiled based on calculated data [3, 4, 5, 6].

The impact of the crisis is most clearly traced in the dynamics of budget revenues from the hydrocarbon sector. As shown in Table 3, even a short-term decline in prices had significant fiscal consequences. It is important to note that during this period the share of fuel exports in Oman's total merchandise export structure exceeded 80%, making the economy almost entirely dependent on fluctuations in global hydrocarbon prices.

**Table 3. Dynamics of Oil Prices and Oman's Budget Oil and Gas Revenues (2000–2002)**

| No. | Year | Average Oil Price (Oman Crude), $/barrel | Budget Oil and Gas Revenues, million OMR | Change from Previous Year, % |
|---|---|---|---|---|
| 1 | 2000 | 26.5 | ~2,850 | +18 |
| 2 | 2001 | 19.2 | ~2,100 | –26 |
| 3 | 2002 | 23.8 | ~2,400 | +14 |

**Note:** data for 2001 are of an estimated nature, reconstructed based on statistical data from the Central Bank of Oman on oil production volumes, IEA average prices, and the tax regime in effect during that period [4, 6, 7].

**Lesson for Oman:** The absence of a budgetary "safety cushion" in the event of a commodity shock multiplies the negative impact on the real sector, even if the shock itself is moderate by historical standards.

**Qatar's Response: Gas Buffer and Long-term Contracts**

Qatar, whose economy had already begun shifting from oil to gas during that period, found itself in a unique position. Although LNG prices also faced pressure, long-term supply contracts to Asia, indexed to oil prices with fixed formulas, ensured a more stable revenue flow compared to spot oil sales [5, 10]. This allowed Qatar to cushion the blow: the decline in hydrocarbon revenues is estimated at ~–18%, the lowest figure among major GCC exporters. Moreover, the dot-com crisis accelerated investments in expanding LNG production capacity (the QatarGas II project), as global energy demand remained high in the long term.

**Qatar's Lesson:** Diversification within the hydrocarbon sector (transition from oil to gas) combined with long-term contracts can create an additional buffer against price shocks and allows maintaining high growth rates even during a global downturn.

**United Arab Emirates' Response: Diversification as a Survival Strategy**

Unlike Oman, the UAE economy demonstrated high resilience due to successful diversification. Despite an identical decline in global oil prices (~–27%), the reduction in UAE budget revenues from oil and gas was only ~–20%, which was significantly lower than in Oman (–26%) or Kuwait (–24%) [1, 2, 5, 6, 7]. The shock's impact was effectively cushioned by the economic structure, where the non-oil sector's share already exceeded 60% in 2001, and accumulated sovereign wealth fund reserves allowed compensating for lost revenues without sharp cuts in government spending. The key role was played by the dual emirate model:

- Abu Dhabi (the oil center) cushioned the blow through currency reserves and support for basic infrastructure projects;
- Dubai (the diversified hub) continued aggressive expansion into logistics, tourism, real estate, and financial services, generating value added independently of oil prices. This expansion, partially financed through transfers from Abu Dhabi, underscored the advantages of the federal model. As a result, the UAE's real GDP growth rate in 2001–2002 maintained stable indicators (~3.1–3.5%), which was more than 4 times higher than Oman's comparable figure (0.7%). Moreover, the 2001 crisis became a catalyst for accelerating diversification programs in Dubai: the launch of large-scale projects (Palm Jumeirah, Dubai Internet City, expansion of Jebel Ali port) allowed the emirate not only to survive the shock but also to lay the foundation for the subsequent economic boom of 2003–2008.

**Lesson for the Region:** The UAE experience clearly demonstrated that structural diversification is not just an abstract goal, but a concrete mechanism of macroeconomic resilience capable of protecting the economy even with a sharp decline in commodity revenues.

**Saudi Arabia's Response: Using Reserves to Protect Social Commitments**

Saudi Arabia, the largest oil exporter in the GCC, recorded a decline in oil and gas revenues of approximately –22% in 2001 [6, 8]. The decline in global prices coincided with a period of moderate reforms; however, the Kingdom had substantial

foreign exchange reserves accumulated in previous years. These funds were directed toward maintaining basic infrastructure projects and financing social commitments, which allowed real GDP to avoid recession (growth of about 1.0% in 2001–2002) [5, 10]. At the same time, the 2001 crisis became one of the signals that initiated discussions on the need for economic diversification, which subsequently culminated in more ambitious programs (including Saudi Vision 2030).

**Saudi Arabia's Lesson:** Even in the absence of sophisticated stabilization mechanisms, a significant volume of international reserves can serve as a temporary buffer, allowing the avoidance of sharp budget contraction and social tension.

### 1.3. Comparative Analysis of GCC Countries' Reactions to the 2001 Shock

The vulnerability of GCC economies to the "dot-com bubble" crisis was heterogeneous and directly dependent on the degree of economic diversification, the presence of sovereign wealth funds, and the structure of the export basket. For clarity, the region's countries are ranked in Table 4 by the degree of budget revenue decline in 2001 (from greatest losses to least).

**Table 4. Comparative Response of GCC Countries to the "Dot-com Bubble" Crisis (2001–2002)**

| No. | Country | Oil Price Decline, % | Budget Oil and Gas Revenue Decline (estimate) % | Impact on Real GDP, % | Key Policy Response and Resilience Factors |
|---|---|---|---|---|---|
| 1 | Oman | ~–27 | ~–26 | Growth slowdown to 0.7 (2002) | Absence of a stabilization fund forced strict spending cuts [4, 6]. |

| | | | | | |
|---|---|---|---|---|---|
| 2 | Kuwait | ~–27 | ~–24 | Maintained growth (~1.5) | Withdrawals from the Future Generations Fund protected the real sector [7, 9]. |
| 3 | Saudi Arabia | ~–27 | ~–22 | Moderate slowdown (~1.0) | Use of foreign exchange reserves to support infrastructure and social commitments [8, 10]. |
| 4 | UAE | ~–27 | ~–20 | Stable growth (~3.5) | Balance between Abu Dhabi's oil reserves and Dubai's diversification [1, 5, 7]. |
| 5 | Qatar | ~–27 | ~–18 | High growth (~6.0) | Long-term LNG contracts and commissioning of North Field capacity [7, 9, 10]. |
| 6 | Bahrain | ~–27 | ~–15 | Stable growth (~4.0) | Low oil dependence, reliance on the financial sector and Alba [6, 8, 11]. |

**Note:** data on budget revenue decline for 2001 for some countries are of an estimated nature, reconstructed based on historical IMF and World Bank reports.

**Sources:** compiled based on calculated data [1, 2, 3, 4, 5, 6, 7, 8, 9, 10].

**Analytical Interpretation of Comparative Table Data**

1. **Highest Vulnerability (Oman):** Oman suffered the most severe consequences in real terms. Lacking a large sovereign fund at that time and having a less diversified economy than the UAE or Bahrain, the Sultanate was forced to respond to the revenue decline with direct cuts in government spending [4, 6].
2. **The "Safety Cushion" Effect (Kuwait and Saudi Arabia):** Despite a comparable decline in oil prices and revenues to Oman, these countries avoided recession. Kuwait demonstrated a classic example of how a stabilization mechanism works through the Future Generations Fund [7, 9].
3. **Structural Resilience (Qatar, UAE, Bahrain):** These countries showed the highest resistance. In Qatar's case, the gas sector with long-term contracts played a decisive role [10]. Bahrain, with the smallest hydrocarbon reserves, relied on the banking sector and industry [11]. The UAE successfully balanced between Abu Dhabi's fiscal support and Dubai's non-oil growth [1, 5].

**Lessons for the Future and Modern Parallels**

The experience of 2000–2002 remains relevant for GCC countries today, in the context of the energy transition and new market volatility. Key lessons include:

- **The necessity of accelerating diversification:** Programs such as Oman Vision 2040 and Saudi Vision 2030 are a direct consequence of lessons learned from past crises;
- **The importance of institutional frameworks:** The creation and formalization of rules for the use of sovereign wealth funds (as in Kuwait) are critically important for fiscal discipline;
- **The flexibility of export policy:** A combination of long-term contracts (for stability) and spot sales (for capturing premiums) can optimize revenues.

**Analytical Conclusions from the Comparative Analysis**

The conducted analysis of GCC countries' response to the "dot-com bubble" crisis (2000–2002) allows us to formulate the following key conclusions:

1. **Heterogeneity of the region's vulnerability.** The crisis clearly demonstrated that GCC countries are not a monolith. Despite an identical external oil price

shock (–27%), the consequences for the real sector differed radically: from a sharp slowdown in Oman (to 0.7%) to maintaining high expansion rates in Qatar (6.0%) and Bahrain (4.0%) [5, 6];

2. **The decisive role of fiscal buffers.** The comparison of Oman and Kuwait is a textbook example of the importance of sovereign wealth funds. With a comparable decline in oil and gas revenues, Kuwait avoided recession thanks to withdrawals from the Future Generations Fund, whereas the absence of a similar mechanism in Oman forced the government to implement strict spending cuts [7, 9];
3. **The effectiveness of structural diversification.** Countries that had already developed non-oil sectors by 2000 showed the highest resilience. Bahrain (financial services and aluminum) and the UAE (logistics, tourism, real estate) were able to compensate for lost oil revenues through alternative sources of value added [8, 10];
4. **The strategic immunity of the gas sector.** Qatar's high growth rates during the crisis proved that long-term LNG supply contracts and physical capacity expansion create a reliable macroeconomic buffer, weakly correlated with short-term oil price volatility [7, 9];
5. **The catalytic effect for macroeconomic reforms.** The recognition of budget vulnerability to exogenous shocks in 2001–2002 became a historical trigger for the region. It was this experience that laid the foundation for the subsequent creation of stabilization funds and the launch of structural transformation programs such as Saudi Vision 2030 and Oman Vision 2040 [8, 10].

## CONCLUSION

The "dot-com bubble" crisis of 2000–2002, being technological and financial in origin as a shock in the United States, had a significant but differentiated impact on the economies of GCC countries through the oil price channel. It clearly demonstrated that the degree of vulnerability of an oil-dependent economy is determined not so much by the scale of the external shock, but by the presence of internal stabilization

mechanisms: sovereign wealth funds, a diversified economic structure, and strategic contracts. For Oman, this crisis became a painful but necessary lesson, pushing the country toward the creation of financial buffers and diversification planning. For the entire region, it served as a reminder of the fundamental need to reduce dependence on cyclical hydrocarbon revenues.

In concluding this analysis, it is appropriate to cite the assessment provided by the International Monetary Fund at the height of reflection on the events that transpired. The March 2002 *Global Financial Stability Report* describes how the crisis impacted the real economy:

> *"The adjustment in financial markets and the real economy following the bursting of the bubble continues to shape developments. This is reflected in the hesitant and uneven pace of global economic growth and corporations' reluctance to increase capital expenditures"* [12].

A bit later, in September 2002, the IMF offered a more pointed characterization of the causes of this collapse:

> *"Market adjustments occurred against the backdrop of the collapse of the bubble in telecommunications, media, and technology companies—a collapse that exposed a culture of reckless optimism, and at times even greed, among many buyers, sellers, and intermediaries"* [13].

These conclusions accurately capture the global paralysis that set in after the crisis. Yet it was precisely this lesson—the lesson of "reckless optimism" and the subsequent contraction of investment—that was absorbed with unprecedented clarity, but in a different context. While the global economy demonstrated hesitancy, the Gulf countries turned someone else's mistake into their own long-term survival strategy. That is why, over the next two decades, the region made a strategic bet on the creation of fiscal buffers, sovereign wealth funds, and structural diversification programs.

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